

Lecture 2 Introduction to Linux Notes

1. What is an Operating System?

An operating system is the fundamental software that enables a user to utilize computer hardware by providing essential tools. It is the face of the computer.

2. What is a kernel?

The kernel is the brain of the computer. It manages system hardware, memory allocation, CPU time and program to program interaction.

3. Which other parts aside from the kernel identify an OS?

A command-line shell which is a program that provides a text-based interface. It allows users to enter commands that the shell passes to the kernel for execution.

4. What is Linux?

Linux is a free, open source and highly customizable Unix-like operating system that is the foundation for internet servers and computing devices.

5. What is a Linux distribution?

Linux Kernel, Core Unix Tools, Supplemental softwares, Start up Scripts and an installer make up the Linux Distribution. The components are Applications, Graphical Desktop Environment, Daemons, Shells, Linux Kernel and Hardware. It is a ready to use version of the Linux operating system and most distributes software packages.

6. List at least 4 Linux characteristics:

Linux is an open source, free of charge, with a system that is highly customizable while also offering broad compatibility with almost any processor architecture.

7. What is Debian?

Debian is a community driven project for free software known as the grandfather of Linux distributions. It serves the base for other systems as well.

8. List and define the different types of licensing agreements

Open Source-It may apply a fee or be free which a source code is distributed. Closed Software-The user is restricted from modifying the code where the source code is not distributed. Freeware-Free of charge but source code is not available. Shareware-Free on a trial basis. Free Software-Source code is distributed.

9. What is Free Software? Define the 4 freedoms.

Freedom 0: Use the software for any purpose Freedom 1: Examine the Source code and modify it as you see fit. Freedom 2: Redistribute the software Freedom 3: Redistribute your modified software.

10. What is virtualization?

Virtualization is using a software to run multiple independent systems at once.